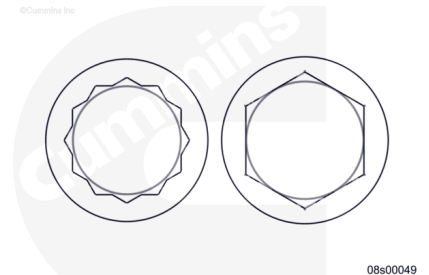


Trainings ▾

General Information

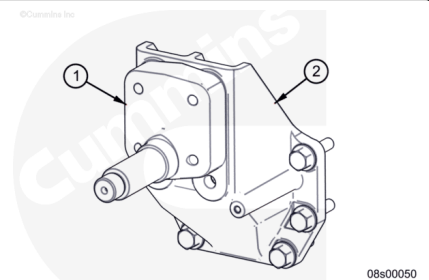
There are two different fan hub shafts available. The shafts can be identified by the difference in mounting capscrew hole size. The mounting capscrew holes will be either 5/8 inch or 3/4 inch thread size.

The different shafts can be identified on engine by the capscrew head shape. The 5/8 inch fan hub shaft mounting capscrew has a twelve point head and the 3/4 inch fan hub shaft mounting capscrew has a six point head.



Due to space constraints, the 3/4 inch mounting capscrew fan hub shaft (1) **must** be removed from the engine as an assembly with the fan hub support bracket (2).

The 3/4 inch mounting capscrew fan hub shaft (1) **must** be assembled to the fan hub support bracket (2) off the engine and installed on to the engine as an assembly.



Preparatory Steps

⚠ WARNING ⚠

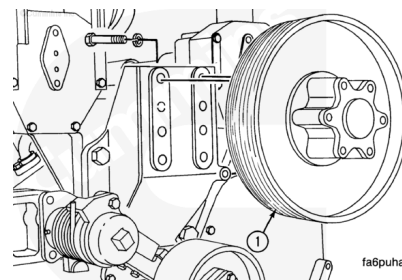
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the cooling fan. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html)
- Remove the cooling fan drive belt. Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Make a note of the holes that are used for the fan bracket. The holes that are used determine the different fan center arrangements.

5/8 Inch Mounting Capscrews

- Remove the four capscrews and washers.
- Remove the fan hub (1).

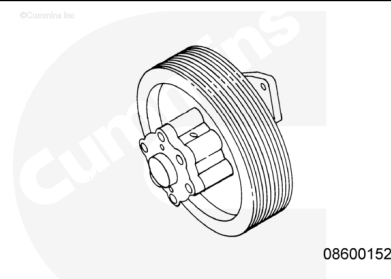
3/4 Inch Mounting Capscrews

- The fan hub shaft and fan hub support bracket **must** be removed from the engine as an assembly and disassembled off the engine. Refer to Procedure 008-035 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html>)

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

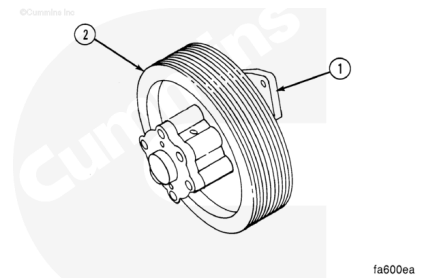
Use solvent to clean the fan hub.

Dry with compressed air.

Check the seal for leaks.

Check the shaft (1) for damage. Check the pulley exterior (2) for damage.

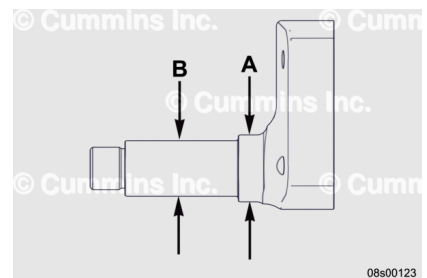
If the shaft or pulley is damaged, the fan hub **must** be rebuilt.



Measure the fan hub shaft.

Dimention A		
mm		in
60.29	MIN	2.374
60.35	MAX	2.376

Dimention B		
mm		in
49.96	MIN	1.9670
49.98	MAX	1.9678

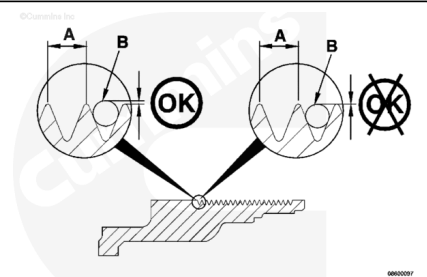


If the clearance is **not** within the specifications, the fan hub **must** be replaced.

Poly Vee pulley groove wear can be measured by placing a gauge ball or drill bit in the pulley groove and checking the protrusion with a straight edge. If the gauge ball or drill bit is below the pulley peaks, the pulley **must** be replaced.

All KV engines use either a K or L section pulley belt.

To determine which gauge ball or drill bit can be used, measure the A dimension and compare to the dimensions of K and L section pulleys.



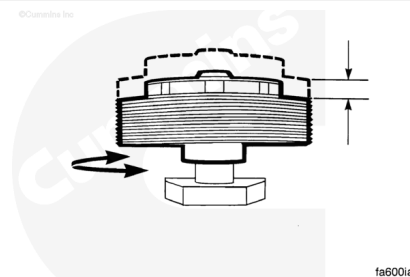
Poly Vee - Pulley Reuse Guidelines		
Belt Type	A	B
K-Section	3.56 mm [0.140 in]	3 mm [0.12 in] gauge ball or a number 31 drill bit

Poly Vee - Pulley Reuse Guidelines

Belt Type	A	B
L-Section	4.70 mm [0.185 in]	4 mm [0.16 in] gauge ball or a number 20 drill bit

Check the clearance.

Bearing End Clearance		
mm		in
0.03	MIN	0.001
0.15	MAX	0.006



If the clearance is **not** within the specifications, the fan hub **must** be rebuilt.

Rotate the pulley to check for rough or damaged bearings.

If the fan hub does **not** rotate freely, the fan hub **must** be rebuilt.

Remove the pipe plugs (3), if installed. Install a grease fitting in one of the holes.

Use high performance, general purpose industrial-type grease (NLGI Grade Number 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

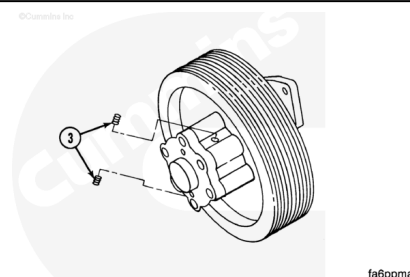
If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Use a grease gun and pump grease into the hub until it begins to come out of the open plug hole.

Remove the grease fitting.

Install the pipe plugs (3).

Torque Value: 15 n•m [133 in-lb]



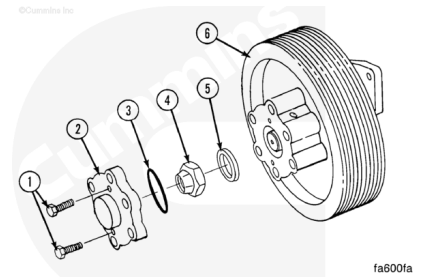
Disassemble

Remove the following parts from the pulley (6).

1. Capscrews

2. Fan pilot
3. O-ring
4. Locknut
5. Plain hardened washer.

Discard the o-ring.

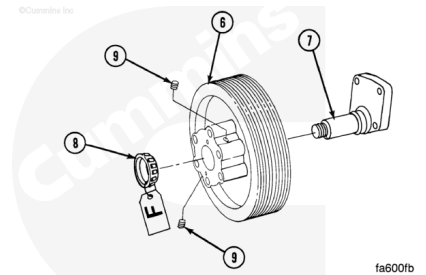


Note : If force is required, use an arbor press. Support the pulley (6) and push on the small end of the shaft (7).

Remove the pulley assembly (6) from the shaft (7).

Remove and tag the front roller bearing (8) for future identification.

Remove the pipe plugs (9), if installed.



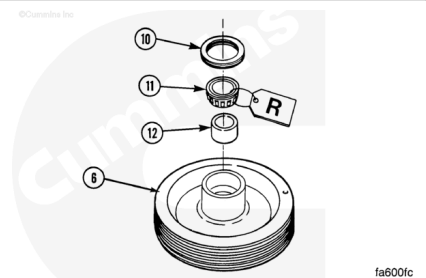
Turn the pulley assembly (6) over.

Remove the oil seal (10).

Remove the rear roller bearing (11) and tag for future identification.

Remove the inner bearing spacer (12).

Note : Do not remove the outer bearing spacer.



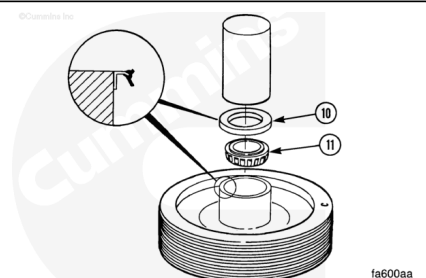
Assemble

⚠ CAUTION ⚠

Do not lubricate the seal surface of the shaft. The seal and seal surface must be clean and dry. Failure to do so can result in seal failure.

Note : Excessive grease will affect the end float reading.

Use high performance, general purpose industrial-type grease (NLGI Grade Number 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.



If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Lightly grease the bearing (11) with grease.

Apply Loctite® 675, or equivalent, to the outside diameter of the seal.

Install the bearing in the hub.

Install the seal (10). Use a mandrel with the flat unseamed face toward the shoulder of the driver and press the seal in until it bottoms.

Install the fan hub seal end first over the shaft and make sure no damage occurs to the seal.

The rear bearing **must** slide easily on the shaft. If it does **not**, check for chips or burrs.

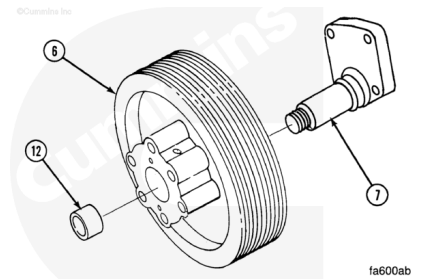
Install the inner bearing spacer (12). The following spacers are available to provide correct end clearance when the fan hub has been assembled.

The following listing is for two spacer style fan hubs:

- 3627681 - 63.906 mm [2.5160 in] length
- 3627682 - 63.995 mm [2.5195 in] length
- 3627683 - 64.084 mm [2.5230 in] length
- 3627684 - 64.173 mm [2.5265 in] length.

The following listing is for other fan hubs:

- 3627677 - 51.562 mm [2.0300 in] length
- 3627678 - 51.689 mm [2.0350 in] length
- 3627679 - 51.816 mm [2.0400 in] length
- 3627680 - 51.943 mm [2.0450 in] length
- 3628830 - 52.019 mm [2.0480 in] length.

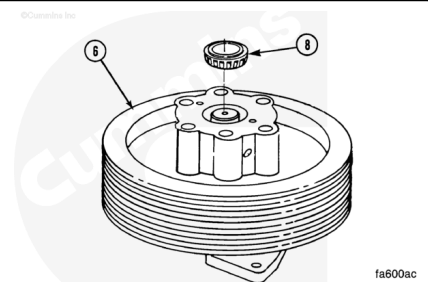


Use high performance, general purpose industrial-type grease (NLGI Grade Number 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Lightly grease the front bearing (8).

Install the bearing on the pulley (6).



The bearing **must** slide easily over the shaft. If it does **not**, check for chips and burrs.

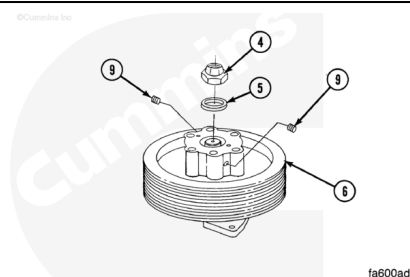
⚠ CAUTION ⚠

Remove all grease from the threads on the shaft before installing the locknut. Grease on the threads reduces the torque retention of the locknut.

Install the hardened washer (5).

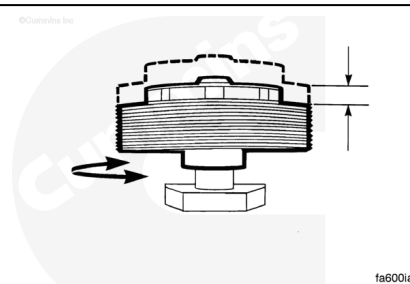
Install the lock nut (4).

Torque Value: 610 n•m [450 ft-lb]



Use a dial indicator to measure the bearing end clearance.

Bearing End Clearance		
mm		in
0.025	MIN	0.001
0.150	MAX	0.006



If the clearance is **not** within specifications, rebuild the fan hub with a suitable size spacer. Check the bearing end clearance.

⚠ CAUTION ⚠

Do not use more grease than specified. Too much grease will cause excessive heating and failure.

Use high performance, general purpose industrial-type grease (NLGI Grade Number 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

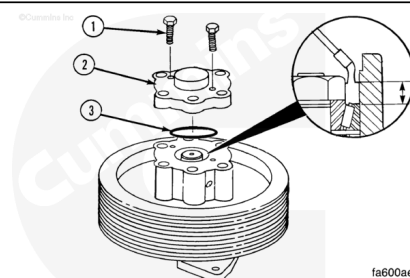
Fill the cavity with 120 cc of grease. Grease the inner race, install the spacer and inner race.

Install the hardened washer. Install the lock nut.

Check the bearing end clearance.

Install the o-ring (3) on the fan pilot (2).

Lubricate the o-ring with vegetable oil.



Install the fan pilot and o-ring.

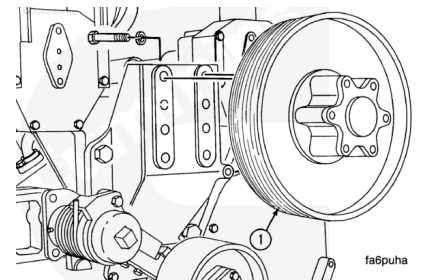
Install the capscrews (1).

Torque Value: 20 n•m [177 in-lb]

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



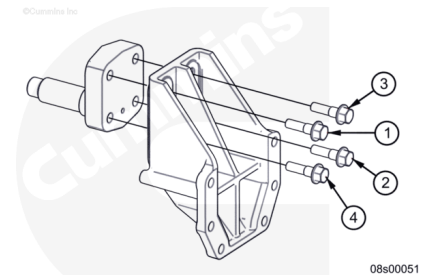
5/8 Inch Mounting Capscrews

- Install the fan hub, four capscrews, and lock washers.
- Tighten the capscrews.

Torque Value: 280 n•m [207 ft-lb]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



3/4 Inch Mounting Capscrews

- Use the four capscrews to assemble the fan hub shaft to the fan hub support bracket.
- Tighten the capscrews in the below sequence.

Torque Value:

1. 100 n•m [74 ft-lb]

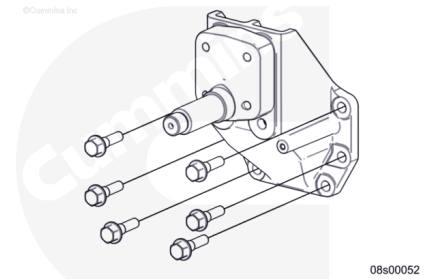
2. 360 n•m [266 ft-lb]

Install the fan hub shaft and fan hub support assembly to the engine. Refer to Procedure 008-035 in Section 8.

(/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html)

Note : The fan hub pulley assembly can be assembled to the shaft once the shaft assembly is installed to the

engine. This will reduce the weight of the fan hub shaft assembly during installation.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the fan belt. Refer to Procedure 008-002 in Section 8.
(</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Install the cooling fan. Refer to Procedure 008-040 in Section 8.
(</qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html>)
- Start and operate the engine and check for proper operation.